

Diabetic microvascular injury

TYPE B | Separate leakage, neuronal ischaemia and neovascularisation.

Capillary wall injury

From supported capillary to leaking sac

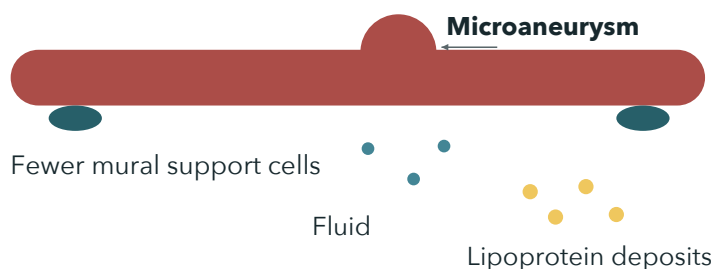
NORMAL WALL SUPPORT

Endothelial tube



Pericytes along capillary wall

DIABETIC CAPILLARY INJURY



Barrier injury permits leakage into the retina

Pericyte loss contributes to microaneurysm formation alongside other vascular injury. Microaneurysms alone do not define proliferative disease.

Leakage leaves two kinds of evidence

Observation	Why it appears
Microaneurysm	Focal damaged capillary dilation
Hard exudate	Leaked lipoprotein-rich material deposited within retina
Macular oedema	Fluid leakage thickens the central retina
Dot/blot haemorrhage	Blood collects in deeper retinal tissue

Central blur does not require new vessels

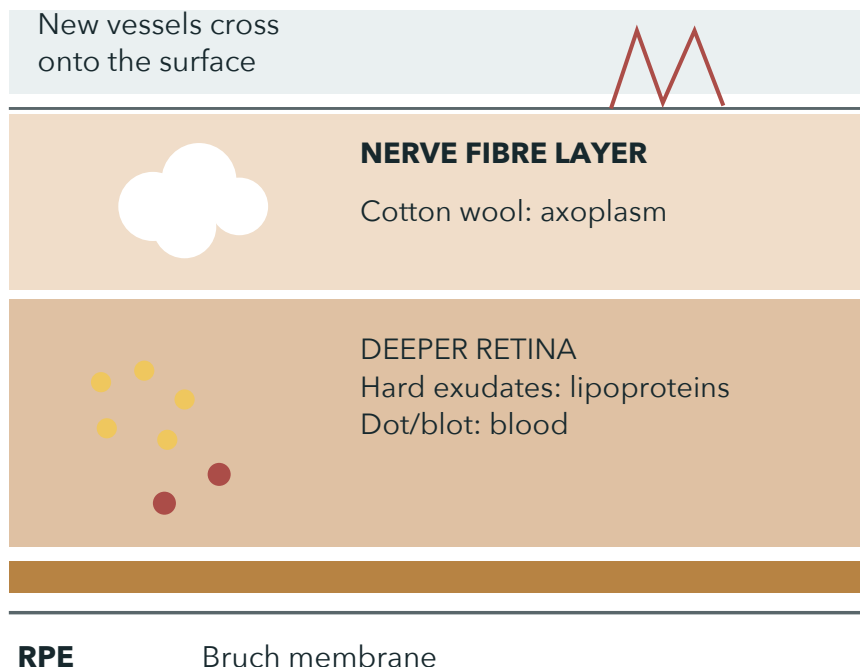
Diabetic macular oedema can occur without proliferative surface neovascularisation. Central capillary ischaemia can also impair vision, so leakage and non-perfusion should be distinguished.

Sources ADA: Human microaneurysm and pericyte analysis | Cusick et al.: Hard exudate histopathology | Lobo et al.: Diabetic leakage and thickening

Recognise the lesion and its plane

Retinal compartment atlas

VITREOUS / RETINAL SURFACE



Drusen lie BELOW RPE, not in the above layers

Cotton wool spots contain accumulated axoplasmic material in superficial ganglion axons. New vessels on the disc or retinal surface establish proliferative diabetic retinopathy.

Two injury paths

Leakage	Non-perfusion
Barrier dysfunction	Insufficient retinal oxygen supply
Oedema + hard exudates	Ischaemic neuronal injury; VEGF signal
Macular fluid can reduce acuity	Fragile new vessels can bleed
Does not establish proliferation	Surface neovascularisation establishes proliferation

Predict the complication

A retinal surface fibrovascular membrane can contract and cause tractional detachment. An iris/angle membrane can obstruct aqueous drainage and cause neovascular glaucoma; these occur in different compartments.

Sources McLeod: Axoplasmic transport and cotton wool spots | NEI: Diabetic retinopathy | Cusick et al.: Hard exudate histopathology